250 results for “(survey OR review) AND (wildfire OR “wild fire” OR “wildland fire” OR “wild land fire” OR “forest fire” OR bushfire) AND (propagation OR spread OR behavior OR dynamics OR trajectory) AND (prediction OR forecast OR simulation OR model OR management)”

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Coupling wildfire spread and erosion models to quantify post-fire erosion before and after fuel treatments

M. Salis | L. Giudice | +8 authors | B. Arca | Environmental Science ·

International journal of wildland fire · 8 October 2019

Wildfires are known to change post-**fire** watershed conditions such that hillslopes can become prone to increased erosion and sediment delivery. In this work, we coupled **wildfire spread** and erosion… Expand

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Transforming Wildfire Detection and Prediction Using New and Underused Sensor and Data Sources Integrated with Modeling

J. Coen | W. Schroeder | S. Rudlosky

Environmental Science, Engineering ·
Handbook of Dynamic Data Driven Applications… · 2018

TLDR The use of additional datasets from satellites with **fire** detection algorithms **and** adjacent non-utilized passes of VIIRS to enhance **simulations** of the 2015 Canyon Creek Complex improved **fire** growth **forecast and** compensated where standard, scheduled observations were missing **or** obscured by clouds. Expand

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Dynamic simulation of fire propagation in forests and rangelands using a GIS-based cellular automata model

N. M. Gharakhanlou | N. Hooshangi

Environmental Science, Geography · International journal of wildland fire ·
25 June 2021

TLDR The proposed **model** considers the most effective spatial **and** temporal drivers of **wildfire propagation**, including wind speed **and** direction, type **and** density of vegetation, **and** topographic conditions, to assist **fire** managers in predicting **fire propagation** to better control **wildfires**. Expand

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Establishing Propagation Nodes as a Basis for Preventing Large Wildfires: The Proposed Methodology

Raúl Quílez L. Valbuena Jordi Vendrell Kathleen Uytewaal J. Ramírez

Environmental Science · Frontiers in Forests and Global Change · 16 December 2020

In Spain, traditional **forest fire management** practices have been conducted for many decades, for both prevention- and extinction-oriented purposes. This **management model** has been forced to shift as a… Expand

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Wildfire Emission Modeling : Integrating BlueSky and SMOKE

G. Pouliot T. Pierce W. Benjey S. O'Neill S. Ferguson

Environmental Science · 2005

Atmospheric chemical transport **models** are used to simulate historic meteorological episodes for developing air quality **management** strategies. **Wildland fire** emissions need to be characterized… Expand

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Modelling, Monitoring and Management of Forest Fires II

C. Brebbia G. Perona Environmental Science, Engineering · 26 May 2010

TLDR A volatile organic compounds flammability approach for accelerating **forest fires and** an integrated system for risk **forecasting and** monitoring are presented. Expand

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Validation of FARSITE model in Mediterranean area

B. Arca Environmental Science · 25 October 2005

In the last two decades several deterministic **models** for predicting **fire** behaviour have been developed. One of the most common **fire** growth **model** is the computer program Farsite (Finney, 1998), based… Expand

5 PDF Save Cite

A Forest Fire Prediction Model Based on Cellular Automata and Machine Learning

Xuan Sun Ning Li +6 authors Ibrahim M. Hezam

Environmental Science, Computer Science · IEEE Access · 2024

TLDR Results from the validation process reveal the potential of the FFSBP **model** as a valuable tool in enhancing **forest fire prediction** accuracy **and** facilitating more robust risk mitigation **and fire**fighting strategies, **and** underscore the potential of the FFSBP **model** as a valuable tool in enhancing **forest fire prediction** accuracy. Expand

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Operational Forest-Fire Spread Forecasting Using the WRF-SFIRE Model

Manish P. Kale Srisai Meher +24 authors Partha Sarathi Roy

Environmental Science · Remote Sensing · 6 July 2024

In the present research, the open-source WRF-SFIRE **model** has been used to carry out surface **forest fire spread forecasting** in the North Sikkim region of the Indian Himalayas. Global **forecast** system… Expand

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Utilizing Deep Learning and Spatial Analysis for Accurate Forest Fire Occurrence Forecasting in the Central Region of China

Youbao Guo Quansheng Hai S. Bayarsaikhan

Environmental Science, Computer Science · Forests · 7 August 2024

TLDR Spatial analysis reveals substantial clustering **and** spatial autocorrelation of **fire** points, indicating high-density areas of **forest fire** occurrence, primarily in Hunan **and** Jiangxi provinces, as well as the northeastern region, indicating the need for tailored **fire** prevention **and management** approaches. Expand

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